Ali NCIBI

Computer science researcher-engineer

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Europe, Africa, Asia

Computer science researcher and engineer with hands-on experience building software systems and machine learning models through my PhD and early professional or personal projects. I bring theoretical rigor, practical model-development expertise, and a systems-oriented mindset to deliver data-driven, reproducible, production-ready software solutions.

Example projects

Flux Media Player – Personal Online — Dec. 2025

A media player and library organizer for personal streaming servers, enabling users to stream and manage IPTV content (Xtream and M3U) through a modern web-based interface. See https://flux-media-player.vercel.app/

Massirah CV Builder – Personal Online — Jun. 2025

A beautiful, responsive, multilingual web application that helps users create clean, ATS-optimized resumes, enhanced with AI-powered job matching and smart suggestions to improve and tailor CVs for specific roles. See https://massirah.live/

FlowAR (Flows of Activity Recognition) Framework – PETRUS team, DAVID lab. Versailles, France — 2023 - 2026

A modular, data-driven operational experimental framework designed to support reproducibility and scientific rigor in the development and evaluation of activity recognition systems. Realized as part of PhD research. See https://github.com/ylaxor/flowar

Relex: a deep-learning-based python package for semantic relations extraction – Personal Online, open-sourced — Aug. 2022

Released Relex as an open-source project & python package aimed to provide easy-to-use pipelines for building custom and deep-learning based semantic relation extraction systems. Used PyTorch. See https://github.com/ylaxor/relex

Task-oriented dialog systems and conversational AI – SouthPigalle Paris, France — 2021 - 2022

Designed and implemented an intelligent chat-bot prototype whose role is to answer users' requests in some specific and technical knowledge context, cf. jewelry/watchmaking.

Temporal links extraction between events found in french text – INSERM Clamart, France — 2021

Designed, implemented and validated a supervised deep-learning-based solution for the task of temporal links (ISO-TimeML, TLINK) extraction between medical events (concepts) in a french text.

Style transfer between music accompaniments in MIDI format – LTCI lab., Telecom Paris Palaiseau, France — 2020 - 2021

Applied deep learning techniques to automatic music generation. Tweaked a generative model (Groove2Groove: https://groove2groove.telecom-paris.fr) for the task.

CNN-based registration of 2D brain tumor images – LTCI lab., Telecom Paris Palaiseau, France — 2020

Studied a CNN-based approach for brain tumor image registration. Evaluated different training losses and explored the resulting transformations on examples from the BraTS MICCAI brain tumor dataset.

Arrhythmia detection from ECG data – RISC lab., ENIT Tunis, Tunisia — 2019

Designed and implemented a neural network for arrhythmia detection through the analysis of patient ECG recordings. Evaluated performance on examples from the MIT-BIH Arrhythmia Database.

Network standards for IoT applications in healthcare – RISC lab., ENIT Tunis, Tunisia — 2018

Reviewed and discussed networking trends for IoT in healthcare applications.

Selected publications

Towards Domain-Robust Activity Recognition using Textual Representations of Binary Sensor Events

Ali Ncibi
*18th International Conference on Agents and Artificial Intelligence **(ICAART)**, 2026.*
https://hal.science/hal-05401764v1

FlowAR: A Framework for Data-Driven Development of Human Activity Recognition Systems using Binary S

Ali Ncibi, Luc Bouganim, Philippe Pucheral
*21st International Conference on Distributed Computing in Smart Systems and the Internet of Things **(DCOSS-IoT)**, 2025. https://hal.science/hal-05213603v1*

FlowAR: une plateforme uniformisée pour la reconnaissance des activités humaines à partir de capteur

Ali Ncibi, Luc Bouganim, Philippe Pucheral
*25ème conférence francophone Extraction et Gestion des Connaissances **(EGC)**, 2025.*
https://hal.science/hal-04944889v1

Data-science toolkit

Theory

Descriptive statistics and statistical hypothesis testing, Statistical Machine Learning, Ensemble Learning, Dimension Reduction, Deep Learning, Multi-task Learning, Meta Learning, Self-Supervised Learning, Natural Language Processing, Computer Vision, Graph Mining, Activity Learning,.

Tools & libraries (development, evaluation and deployment)

PyTorch, Tensorflow, Keras, Flair NLP, NLTK, Spacy, Transformers, Pandas, Scikit-learn/network/optim, Matplotlib, Seaborn, Plotly, Jupyter, ONNX, Tensorflow Js, Streamlit, Flask, FastAPI, etc.

Familiar data types and tasks

- Tabular data: Regression, Classification, Clustering.
- Text data: Text Classification, Named Entity Recognition, Part-of-Speech Tagging, Natural Language Understanding, Natural Language Generation, Text Paraphrasing, Question Answering.
- Image data: Image Classification, Object Recognition, Image Captioning, Image Registration.
- Audio data: Music Generation, Style Transfer.
- Sensor data: Activity Discovery, Activity Recognition, Activity Prediction, Dynamic Segmentation

Education

PhD in Computer Science – University of Paris Saclay Mar. 2023 - Feb. 2026
Versailles, France

Focused on the design of structured data-driven pipelines and the evaluation of semantic representation strategies for cross-home human activity recognition from IoT traces.

Master of Science in Data Science – Telecom Paris Palaiseau, France 2019 - 2021

Focused on numerical optimization, statistical machine learning, computer vision, natural language processing, graph mining, programming, and human & social sciences.

Master of Science in Information Technology and Telecommunications 2017 - 2019
– The National Engineering School of Tunis (ENIT) Tunis, TU, Tunisia

Focused on electronics, signal transmission and processing, networking techniques, programming, databases, and system administration.

Master of Science in Mechanical Engineering – Supméca Paris, France 2016 - 2017

Completed the first year of the program before shifting focus to a career in IT and computer science.

Preparatory classes for entrance to tunisian/french Grandes Ecoles engineering schools 2014 - 2016
– Scientific and Technical Studies Preparatory Institute (IPEST)
Al Marsa, TU, Tunisia

Specialized in mathematics and physics.

Computer skills

Programming

- Python
- C/C++
- C#
- R
- JavaScript
- SQL

Computer skills

Web & mobile development

- HTML5 and CSS3
- React
- Node.js
- Vue.js (Nuxt.js)
- Ionic/Capacitor

Productivity & authoring tools

- Latex
- Microsoft Office
- Adobe Creative Suite
- Draw.io

Computer skills

Version control

- Git

Containerization

- Docker

Computer skills

Operating systems

- Linux
- macOS
- Windows

Cloud platforms

- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)
- Heroku
- DigitalOcean
- Supabase, Vercel, Netlify, etc.

Languages

French

Fully functional

English

Good working knowledge

Arabic

Native